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Breathing in Empty Space

Finding Gravity within Quantum Mechanics

Quantum field theory is able to successfully account for every experiment ever performed by human beings. When it comes to describing reality, it's the best approach we have. It's therefore extremely tempting to imagine that future physical theories will be set within the broad paradigm of quantum field theory, or perhaps small variations thereof.

But gravity, at least when it becomes strong, doesn't seem to be well described by quantum field theory. So in this chapter we'll ask whether we can make progress by attacking the problem from a different angle.

Following Feynman, physicists love to remind one another that nobody really understands quantum mechanics. Meanwhile, they have long lamented that nobody understands quantum gravity. Maybe these two lacks of understanding are related. Gravity, which describes the state of spacetime itself rather than just particles or fields moving within spacetime, presents special challenges when we try to describe it in quantum terms. Perhaps that shouldn't be surprising, if we don't think we fully understand quantum mechanics itself. It's possible that thinking about the foundations of quantum theory—in particular, the Many-Worlds perspective that the world is just a wave function, and everything else emerges out of that—will shed new light on how curved spacetime emerges from quantum underpinnings.

Our self-appointed task is one of reverse engineering. Rather than taking classical general relativity and quantizing it, we will try to find gravity within quantum mechanics. That is, we will take the basic ingredients of quantum theory—wave functions, Schrödinger's equation,